

NICHOLAS TOAR TURANGAN

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[Website](#) | [LinkedIn](#)

EDUCATION

University of California, Berkeley | College of Computing, Data Science and Society

Berkeley, California

Bachelor of Arts in Data Science – Applied Mathematics Domain

May 2026

Minor: Statistics

Relevant Coursework : Probability, Time Series, Stochastic Processes, Statistical Inference, Numerical Analysis, Optimization, Machine Learning

EXPERIENCE

University of California, Berkeley, Haas School of Business

Berkeley, California

Research Data Analyst

September 2024 - Present

- Conducted **longitudinal time-series analysis** of Haas' U.S. News, Financial Times, and Poets & Quants ranking data to quantify performance trends and identify the statistical factors driving year-over-year changes.
- Developed an automated **Python pipeline** for data ingestion and cleaning (Google Scholar, Scopus APIs) with duplicate detection by DOI/title; cut data-processing time by **90%** while ensuring reproducibility through audit logs.
- Produced **executive-level forecasts and visualization dashboards** that translated model results into strategic recommendations for the Dean's office and board meetings—pinpointing specific “points lost” categories.

Kebun Wisata Pasirmukti

Bogor, Indonesia

Business Analytics / Revenue Optimization Intern

May 2025 - September 2025

- Built a Python-based revenue and demand analytics workflow using **12+** months of booking data to forecast visitor volume across school trips, family visits, corporate outings, weekends, and holiday periods.
- Consolidated **1,000+** booking, attendance, package, and revenue records into a structured dataset, enabling repeatable analysis of seasonality, average booking size, capacity utilization, and segment-level revenue trends
- Developed scenario analyses for pricing, staffing, and promotional periods, translating demand forecasts and revenue sensitivity analysis into management-facing recommendations for operational planning.

San Jose State University Intelligent Conversational Agents and Neural Networks Lab

San Jose, California

Volunteer Undergraduate Researcher

June 2023 - August 2023

- Implemented and fine-tuned NLP models (**BERT, RoBERTa**) to analyze sentiment analysis around TikTok's potential US ban in 2023, achieving **>90%** accuracy while developing automated data pipelines for processing.
- Conducted **statistical performance analysis** of model outputs using accuracy, F1-score, and confidence calibration metrics to evaluate generalization behavior, bias, and robustness across data segments.

PROJECTS

Atlas Allocation / ETF Portfolio Research & Optimization System

Python (NumPy, pandas, SciPy, scikit-learn, matplotlib, yfinance), Streamlit, TypeScript, Vercel

- Built a systematic ETF allocation research platform for regime-aware portfolio construction, combining alpha signals, market-state logic, checkpointed allocators, and benchmark-aware backtesting.
- Researched 7 frontier phases across state quality, trend strength, re-risking, leadership/crowding, allocator objectives, ML diagnostics, and cross-asset lead-lag behavior.
- Promoted a Phase 5 fragility-guard strategy to production, improving Sharpe **0.884 → 0.948** and max drawdown **-13.98% → -11.60%**, with **2.179 holdout Sharpe**.

TECHNICAL SKILLS

Programming: Python (NumPy, pandas, SciPy), R, SQL (MySQL), Git, Jupyter

Quantitative Methods: Time-series analysis, forecasting, Monte Carlo simulation, optimization, backtesting

Investment Analytics: Portfolio construction, alpha signal research, risk/return analysis, drawdown analysis

Machine Learning: Supervised learning, model validation, regularization, calibration, feature engineering